The APL block model, built step by step

> **QUESTION**
>
> How is the APL block model built, step by step, and what does it predict about blocking APL, before any comparison with the TNT / TNT3c recordings?

INPUTS

- Derived KC claws: `hemibrain_KC_claws_derived.csv.gz`, `malecns_KC_claws_derived.csv.gz` (Report 02)
- Per-KC calyx APL synapse counts: `KC_weights_APL_PN_per_KC.csv` (Report 01)
- Per-glomerulus APL wiring: `odor_weighted/APL_wiring_by_glomerulus.csv` (Report 04)
- Yang's ORN table `ORN_ovlp16_resp_flyave_merge_dropna.csv` (36 glomeruli × 16 odors) and the GH146 means `odor_weighted/GH146_mean_by_glomerulus.csv` (Report 03)

OUTPUTS

- `odor_weighted/APL_block_model_per_odor.csv`, `APL_block_model_odor_pairs.csv`, `APL_block_model_summary.json`, `APL_block_model.png`

SCRIPTS

- `scripts/apl_block_model.py` (the model)
- `scripts/reports/r05_block_model.py` (this report; it re-runs the steps for illustration and checks them against the saved tables)

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This report explains the KC model that all later steps use. Report 07 takes the same model and pins it to the recordings. Here it runs in its **prediction** form, set up before the recordings arrived: 20% of KCs respond to an odor (averaged over odors) with APL blocked, and 10% with APL intact. Every symbol is defined in Table 1, and again where it first appears.

The model in one paragraph

Each KC adds up its claws' glomerular inputs (a weighted average of the glomeruli's odor responses, with weights from the connectome). It responds by how much that sum exceeds a threshold. APL is one cell whose activity, for each odor, is the average response of the KCs. It inhibits every KC by subtracting a fixed multiple of that activity. Because APL's activity depends on the KC responses and the KC responses depend on APL's activity, both are solved together for each odor ("self-consistently"). "Blocked" (TNT) means no APL at all. Later versions replace the uniform APL with connectome measurements of where APL's synapses are.

Table 1. Symbols used in this report.

Symbol	What it stands for	Values, units, range

Symbol	What it stands for	Values, units, range
k	Index of a Kenyon cell (KC) in one mushroom body	$k = 1 \dots N$
N	Number of KCs in the mushroom body (KCs with at least one claw)	hemibrain R 1,793; male CNS L 1,876; male CNS R 1,900
g	Index of an antennal-lobe glomerulus (uniglomerular PN types only)	names such as DM2, VA1v
o	Index of an odor	16 odors (ORN table) or 13 (GH146)
c	Index of a claw (a cluster of a KC's PN synapses, Report 02)	5.6–8 claws per KC
n_c	Number of PN→KC synapses on claw c	synapse count
$g(c)$	Glomerulus of claw c: the glomerulus of the PN that makes most of the claw's PN synapses	
W_{kg}	Input weight: share of KC k's claw PN synapses that sit on claws of glomerulus g	0 to 1; a row sums to the share from uniglomerular PNs (0.96–0.97 on average)
r_{go}	Response of glomerulus g to odor o; negative values set to 0	ORN table: fly-averaged response as provided (ΔF/F-like, units not stated), 0 to 2.06; GH146: mean peak ΔF/F
x_{ko}	Drive of KC k by odor o: weighted average of its glomeruli's responses	same units as r_{go}
θ	KC threshold, one number per mushroom body and input table	set so 20% of KCs respond with APL blocked (Section 5)
y_{ko}	Response of KC k to odor o (how far the drive exceeds threshold plus inhibition; 0 if below)	same units as r_{go}; ≥ 0
A_o	APL activity for odor o: the (weighted) mean KC response	same units as y; one number per odor
a	Subtractive APL gain (how strongly APL activity inhibits a KC), one number per mushroom body and version	dimensionless; set by calibration
b	Divisive APL gain on PN boutons (versions P0, P, L)	dimensionless; set by calibration
u_k	Per-KC inhibition weight: KC k's APL→KC synapses in the calyx ÷ the mean over KCs	mean 1; $u_k = 1$ in the uniform version
v_k	Per-KC drive weight: KC k's KC→APL synapses in the calyx ÷ the mean over KCs	mean 1; $v_k = 1$ in the uniform version
p_g	Bouton weight of glomerulus g: APL→PN synapses per PN→KC synapse of its PNs, scaled to a claw-input-weighted mean of 1	version P (Report 04)
q_g	Claw weight of glomerulus g: APL→KC synapses on its claws per PN→KC synapse on them, scaled the same way	version L (Report 04)
f	Boutons' share of APL's calyx output synapses: APL→PN ÷ (APL→PN + APL→KC)	hemibrain 0.097, male CNS 0.112
s	Common scale of the KC and bouton gains in versions M0 and M	set by calibration
$F_o(A)$	The APL activity the KCs would produce for odor o if APL's activity were A	used to solve for A_o

Symbol	What it stands for	Values, units, range
ϕ_o	Fraction of KCs responding to odor o ($y_{ko} > 0$), also called the coding level	0 to 1
S_{ab}	Cosine similarity of the KC population responses to odors a and b	0 to 1 here (responses are ≥ 0)
ΔS_{ab}	Effect of blocking APL on the similarity of odors a and b: S_{ab} blocked − S_{ab} intact (written Δcos in the figure files)	positive = more similar when blocked
w_g	Total claw input from glomerulus g in the mushroom body, $\sum_k W_{kg}$	used to scale glomerulus weights

1 What the model is

The model is a rate model of the calyx. A rate model describes each cell by one number per odor (its response), not by spikes or a time course. There is one model per mushroom body: hemibrain R, male CNS L and male CNS R. FlyWire and BANC were dropped on 2026-10-21 at Yang's request. Figure 1 shows the circuit and where each symbol acts.

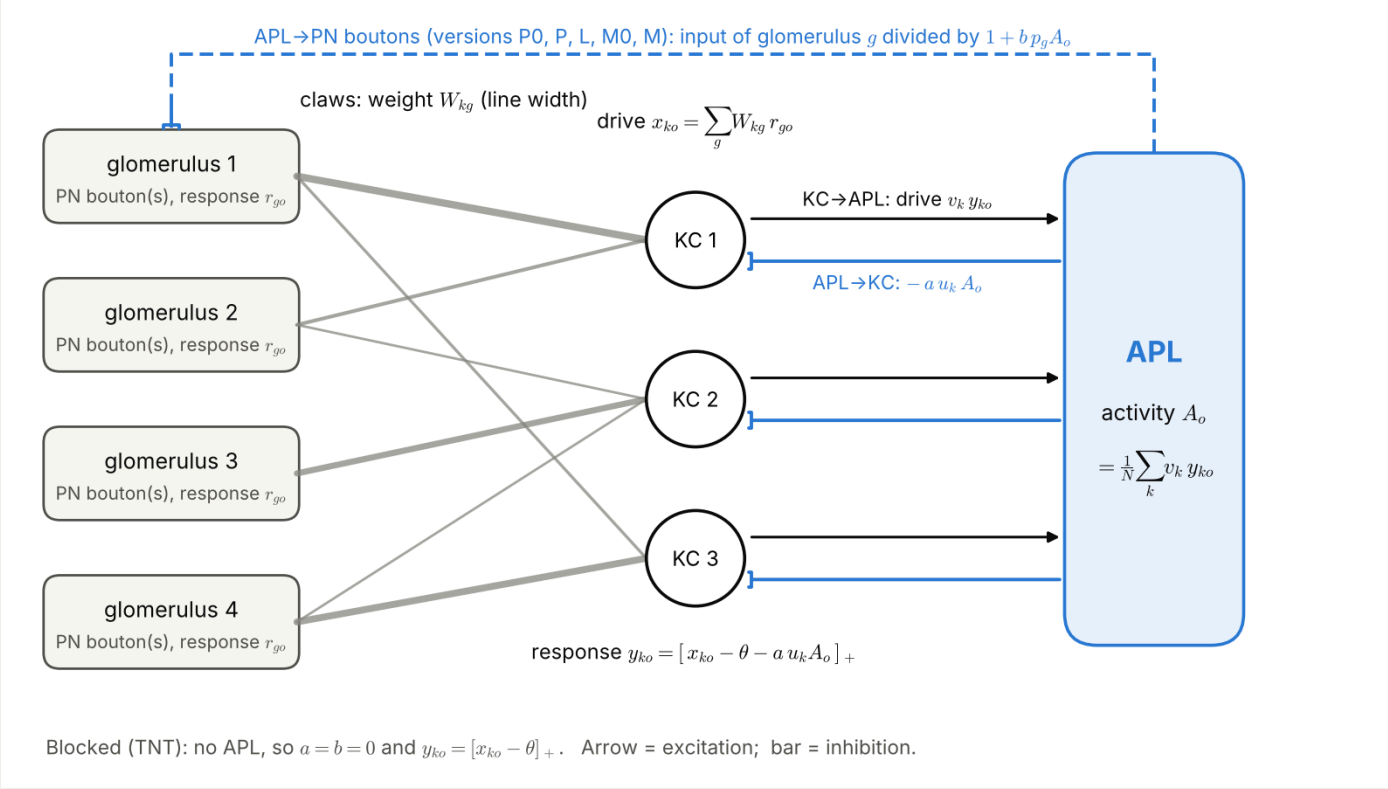


Figure 1. The circuit in the model and where each symbol acts. Glomeruli (left) send PN input to KCs (middle) through claws. Line width shows the input weight W_{kg}, the share of a KC's claw PN synapses that come from each glomerulus (three example KCs). A KC's drive x_{ko} is the W-weighted sum of its glomeruli's responses r_{go}. KCs excite APL (black arrows), and APL's activity A_o is their mean response, each KC weighted by v_k. APL inhibits each KC by subtracting $a\,u_k\,A_o$ (blue bars). In the bouton versions it instead divides the input of glomerulus g by $1 + b\,p_g\,A_o$ (dashed blue path). In the uniform version every u_k, v_k and p_g is 1.

2 Step 1: one model per mushroom body

The hemibrain contains one calyx, the right one (1,793 KCs with claws). The male CNS contains both. Its KCs are split into left and right in two steps. First, each KC's position is the mean of its claw centroids. Second, two-means clustering (k-means with $k = 2$, seed 0) on those positions gives two groups. Each

group is named after the APL that its KCs' on-claw APL synapses mostly contact (APL_L or APL_R). This gives 1,876 KCs on the left and 1,900 on the right (Figure 2). Each mushroom body gets its own W, u, v and threshold, and its own calibrated gains. Results are averaged over the three only at the end.

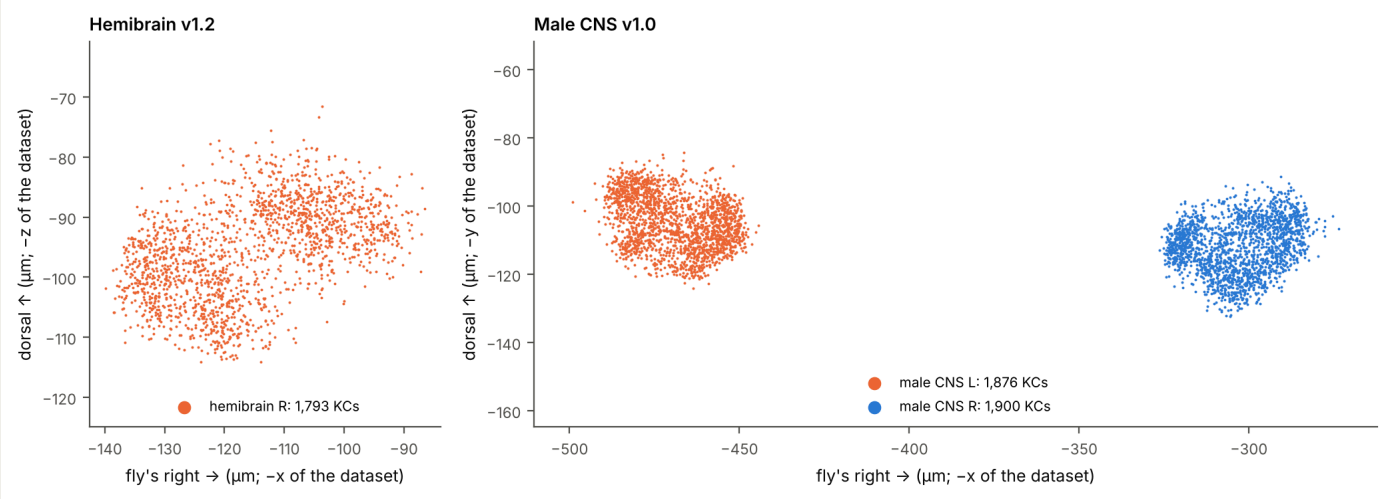


Figure 2. KCs of each mushroom body. Each dot is one KC, placed at the mean of its claw centroids and seen from behind the fly (posterior view: the fly's right on the right, dorsal up). Axes are the dataset's own coordinates in µm, sign-flipped for this view: hemibrain (−x, −z), male CNS (−x, −y). Color shows the mushroom body each KC is assigned to. In the male CNS the two groups come from two-means clustering and are named by the APL their on-claw synapses contact.

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3 Step 2: input weights from the claws

A **claw** is a cluster of one KC's postsynaptic sites from antennal-lobe PNs in the calyx. Sites within 2 µm of each other (single linkage) belong to the same claw. Report 02 describes how claws are found. Each claw is assigned to one glomerulus, the glomerulus of its **main PN** (the PN that makes most of the claw's PN synapses), and all its PN synapses count for that glomerulus. Only uniglomerular PN types count. Claws whose main PN is multiglomerular, bi-glomerular or untyped stay in the denominator but give no input. The input weight is

$$W_{kg} = \frac{\sum_{c \in \text{claws}(k):\, g(c)=g} n_c}{\sum_{c \in \text{claws}(k)} n_c} \tag{1}$$

n_c = PN synapses on claw c; the sum on top runs over KC k's claws of glomerulus g, the sum below over all its claws

Worked example. Table 2 follows one hemibrain αβ KC (body ID 1109223365, type KCab-m) through Equations 1 and 2 for the odor pentyl acetate (pa). Its 6 claws carry 108 PN synapses in total.

Table 2. One KC's claws, its input weights and its drive by pentyl acetate. Each row is a claw. Its share of the KC's claw PN synapses is its contribution to W_{kg}; claws of the same glomerulus add up. The last column is the claw's contribution to the drive. Glomeruli missing from the ORN table count as 0.

claw	main PN type	glomerulus $g(c)$	PN synapses n_c	share = contribution to W_{kg}	$r_{g,\text{pa}}$ (ORN, ≥ 0)	$W \times r$
claw 1	VL2a_adPN	VL2a	30	0.278	0.035	0.010
claw 2	DM1_lPN	DM1	19	0.176	0.791	0.139
claw 3	DM1_lPN	DM1	19	0.176	0.791	0.139

claw	main PN type	glomerulus $g(c)$	PN synapses n_c	share = contribution to W_{kg}	$r_{g,\mathrm{pa}}$ (ORN, ≥ 0)	$W \times r$
claw 4	DM1_lPN	DM1	14	0.130	0.791	0.103
claw 5	V_ilPN	V	13	0.120	not in table (0)	0.000
claw 6	VA4_lPN	VA4	13	0.120	not in table (0)	0.000
sum			108	1.000		**0.391** = $x_{k,\mathrm{pa}}$

The **drive** of KC k by odor o is the W-weighted sum of its glomeruli's responses:

$$x_{ko} = \sum_g W_{kg}\, r_{go} \tag{2}$$

Because the weights of one KC add up to at most 1, x_{ko} is a weighted average of the responses of the KC's glomeruli. Claws of glomeruli outside the response table count as 0. The drive is therefore smaller for KCs whose claws come mostly from unmeasured glomeruli.

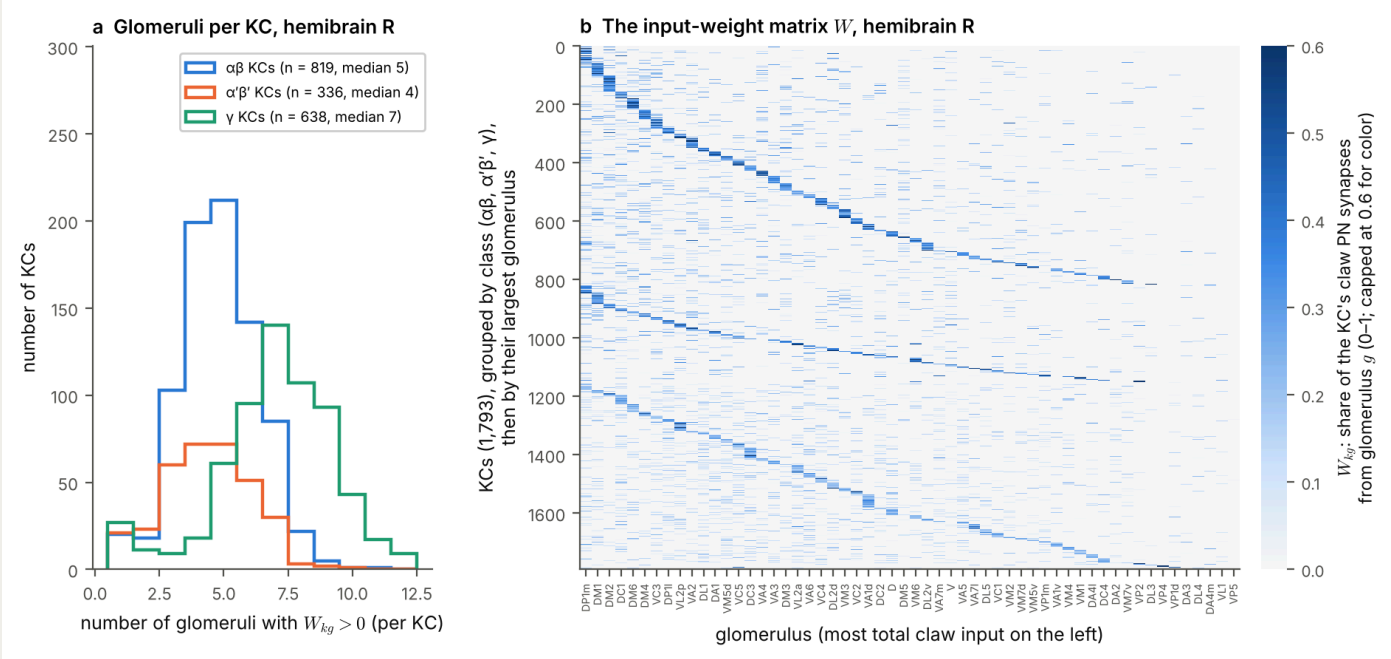


Figure 3. Input weights W_{kg} across all KCs. a, For each KC, the number of glomeruli that give it any input ($W_{kg} > 0$), as a histogram per KC class. **b**, The whole matrix W for hemibrain R. Each row is a KC and each column a glomerulus; color is W_{kg} (white = 0). Values above 0.6 share the darkest color. Every KC samples a few glomeruli, and different KCs sample different combinations.

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4 Step 3: odor input to the glomeruli

Two tables give r_{go}, the response of glomerulus g to odor o. The model takes no PN-level transformation: these numbers go straight into Equation 2. Negative responses (inhibition below baseline) are set to 0.

- **ORN table** (Yang's `ORN_ovlp16_resp_flyave_merge_dropna.csv`): 36 glomeruli × 16 odors, fly-averaged olfactory receptor neuron (ORN) responses. The file does not state units; the values behave like ΔF/F (range −0.19 to 2.07). Glomeruli in this table carry 76%–77% of the KCs' claw PN synapses (mean over KCs, per mushroom body).

- **GH146 table**: mean peak ΔF/F of GH146 PNs (7 flies), 29 glomeruli × 13 odors. -aPine, euc and B-myr are missing, and paa comes from the diagnostics panel at 10^{-5} because it is the only PAA presentation. These glomeruli carry 63%–64% of the claw PN synapses.

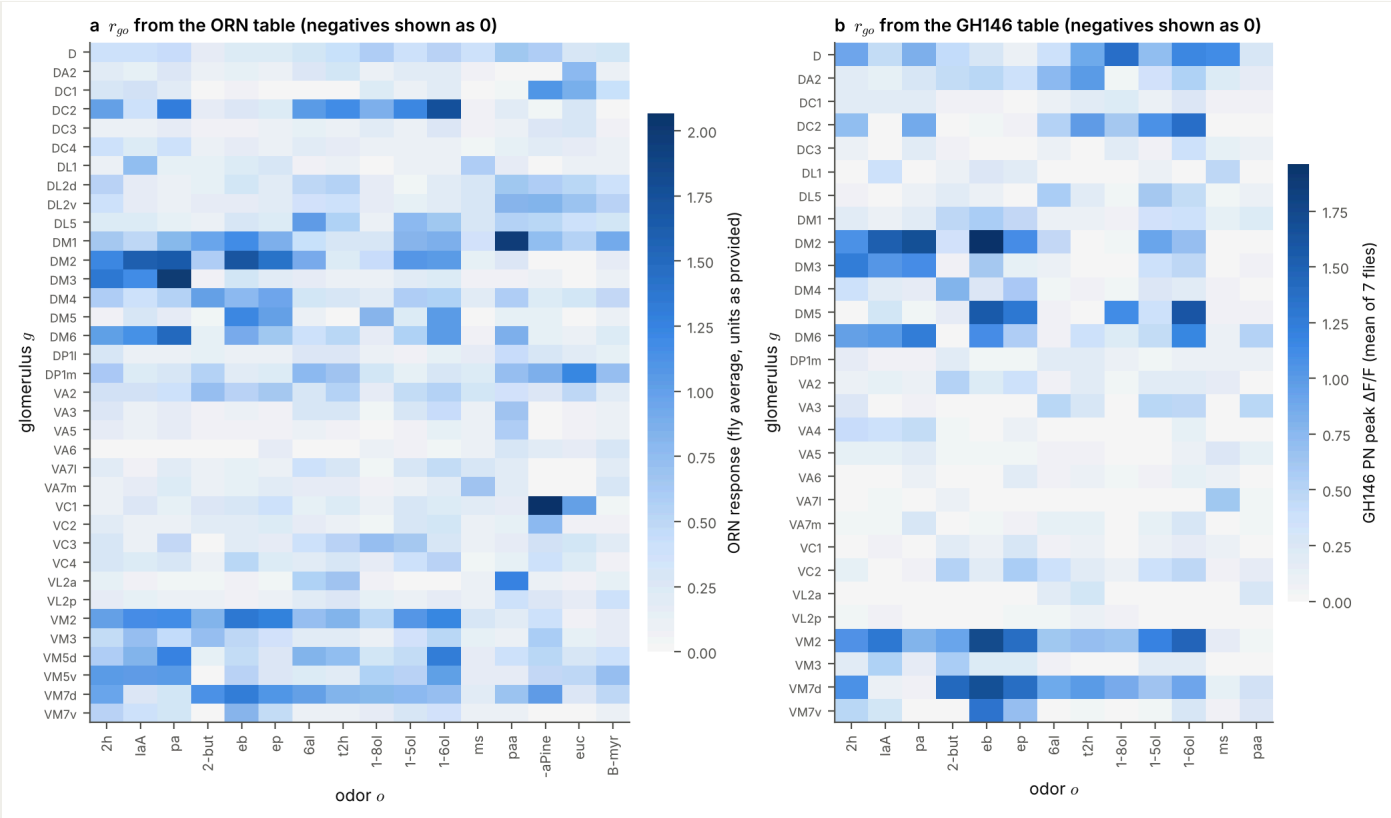


Figure 4. The glomerular responses r_{go} that drive the model. Each cell is one glomerulus's response to one odor after negative values are set to 0. Color runs from 0 (white) to the table's maximum (dark blue), with a separate scale for each panel. **a**, ORN table, 36 glomeruli × 16 odors. **b**, GH146 table, 29 glomeruli × 13 odors. Glomeruli that are missing here give 0 input to the claws that belong to them.

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5 Step 4: KC drive and the threshold with APL blocked

With APL blocked there is no inhibition, so a KC responds by how far its drive exceeds the threshold θ:

$$y_{ko}^{\text{blocked}} = [\,x_{ko} - \theta\,]_+ \qquad \text{where } [z]_+ = \max(z, 0) \tag{3}$$

θ is one number per mushroom body and input table. It is found by bisection so that, averaged over odors, 20% of KCs respond ($y_{ko} > 0$). The bisection keeps an interval $[\theta_{lo}, \theta_{hi}]$, which starts at 0 and the largest drive. At each step it tests the midpoint: if more than 20% respond there, the threshold must be higher, so the midpoint becomes θ_{lo}; otherwise it becomes θ_{hi}. 60 halvings make the interval negligibly small. Individual odors then respond at different levels (Figure 5). Strong odors drive many KCs past the shared threshold, and weak ones drive few.

$$\frac{1}{O}\sum_{o=1}^{O} \phi_o^{\text{blocked}} = 0.20, \qquad \phi_o = \frac{1}{N}\#\{k : y_{ko} > 0\} \tag{4}$$

O = number of odors; ϕ_o = fraction of KCs responding to odor o

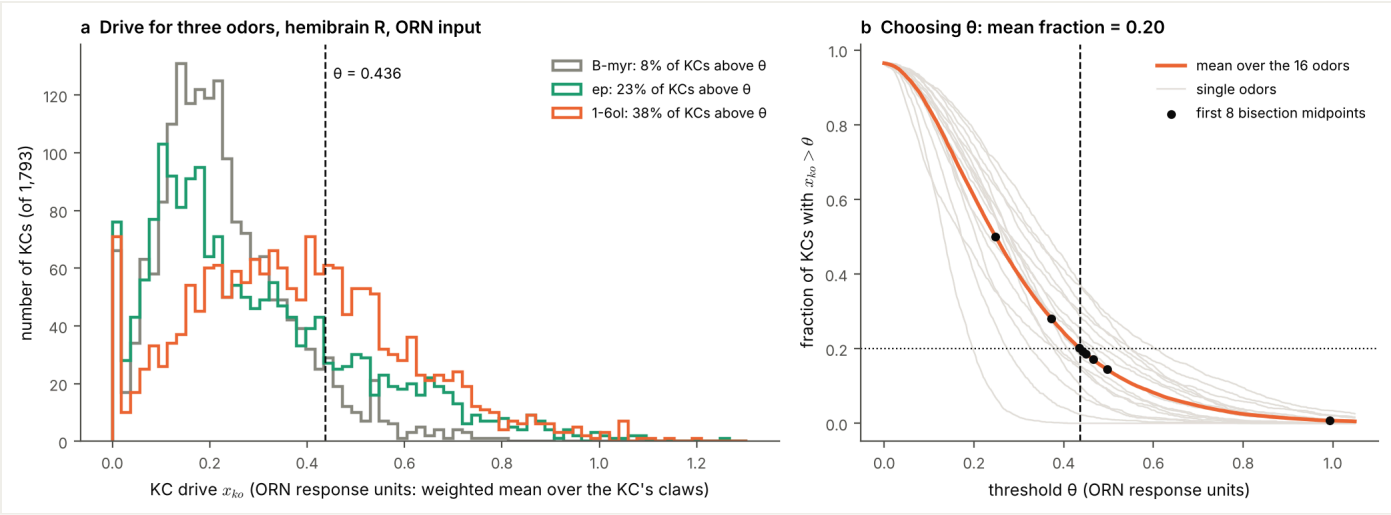


Figure 5. The KC drive and how the threshold θ is set. a, Histograms of the drive x_{ko} over the 1,793 hemibrain KCs for a weak, a middle and the strongest odor (ORN input). The dashed line is the threshold θ. KCs to its right respond when APL is blocked. **b**, The fraction of KCs above a threshold, as the threshold rises: thin grey lines are single odors, and the orange line is their mean. θ is where the orange line crosses 0.20 (dotted). Black dots are the first bisection midpoints as they close in on it.

Table 3. Thresholds. In each case θ makes 20% of KCs respond with APL blocked, averaged over odors. The units are those of the response table.

mushroom body	θ with ORN input	θ with GH146 input
hemibrain R	0.436	0.274
male CNS L	0.451	0.277
male CNS R	0.462	0.297

6 Step 5: APL feedback and what "solved self-consistently" means

With APL intact, each KC is inhibited in proportion to APL's activity for that odor:

$$y_{ko} \;=\; [\,x_{ko} - \theta - a\,u_k\,A_o\,]_+ \tag{5}$$

and APL's activity is the mean response of the KCs, each weighted by how strongly it drives APL:

$$A_o \;=\; \frac{1}{N}\sum_{k=1}^{N} v_k\,y_{ko} \tag{6}$$

In the uniform version (U) every $u_k = v_k = 1$. Every KC then gets the same inhibition aA_o for a given odor, and APL's activity is simply the mean KC response. The gain a is one number per mushroom body and version. It is the same for every odor and every KC. The inhibition still differs between odors, because A_o does.

Why "self-consistently"

A_o appears on both sides. The KC responses (Equation 5) depend on APL's activity, and APL's activity (Equation 6) depends on the KC responses. The model needs the one value of A_o that reproduces itself: the value that, when used to compute the KC responses, gives back the same A_o as their mean. Mathematically this is a fixed point. Physically, it is the steady state in which APL and the KCs have settled.

To find it, define for each odor the function

$$F_o(A) \;=\; \frac{1}{N}\sum_k v_k\,[\,x_{ko} - \theta - a\,u_k\,A\,]_+ \tag{7}$$

the APL activity the KCs would produce if APL's activity were A

and solve $A = F_o(A)$. F_o starts at $F_o(0)$, the mean (weighted) response with no inhibition, and can only fall as A grows, because more inhibition never raises a KC's response. The line A rises from 0. The two cross exactly once, which makes the solution unique (Figure 6). The code finds it by bisection on $[0, F_o(0)]$: if $A > F_o(A)$ at the midpoint, the crossing lies below it; otherwise above. 60 halvings, done for all odors at once (function `fixed_point` in `apl_block_model.py`).

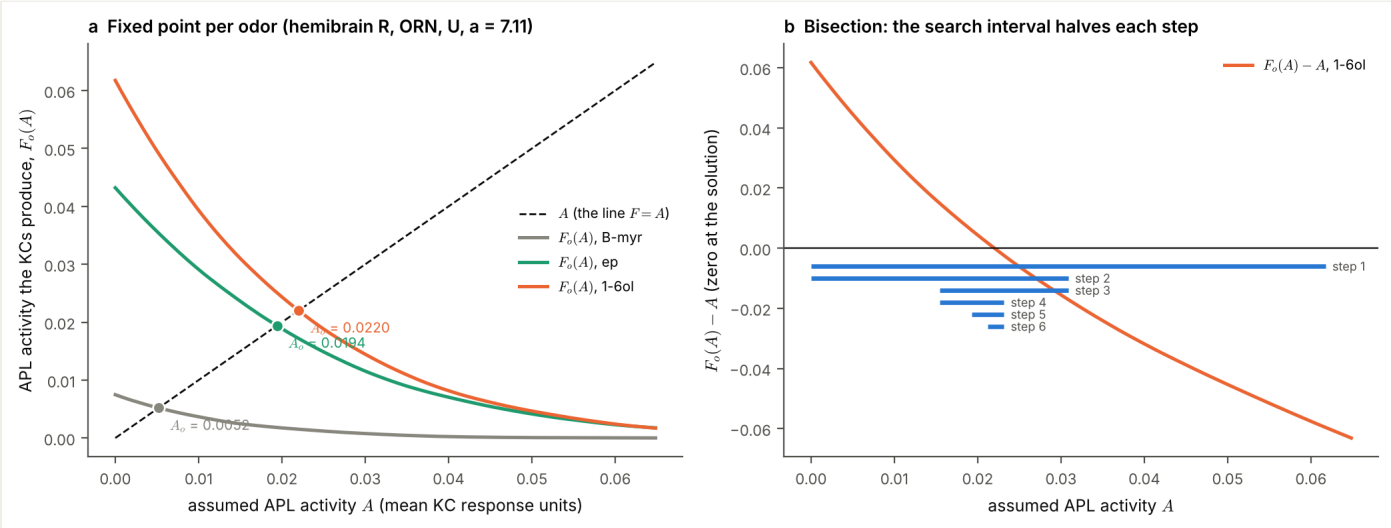


Figure 6. Solving for APL's activity: the fixed point $A_o = F_o(A_o)$. **a**, For three odors (a weak one, a middle one and the strongest; hemibrain R, ORN input, uniform APL at its calibrated gain), the APL activity that the KC responses would produce, $F_o(A)$ (colored), against the assumed activity A. The dashed diagonal is $F = A$, and the dots mark where each curve crosses it: the self-consistent APL activity A_o. Stronger odors cross further out, so they recruit more APL. **b**, The bisection for the strongest odor. Blue bars are the search interval $[A_{lo}, A_{hi}]$ at each step (drawn below zero for visibility). Each step halves the interval around the zero of $F_o(A) - A$.

Table 4. The first bisection steps for 1-6ol (hemibrain R, ORN input, uniform APL). The interval starts at $[0, F_o(0)]$. After 60 steps it has shrunk by a factor of 2^{60}. The final A_o is 0.022012.

step	A_{lo}	A_{hi}	midpoint m	$F_o(m)$	decision
1	0.00000	0.06176	0.03088	0.01378	m > F(m): solution below m, so $A_{hi} = m$
2	0.00000	0.03088	0.01544	0.03038	m ≤ F(m): solution above m, so $A_{lo} = m$
3	0.01544	0.03088	0.02316	0.02073	m > F(m): solution below m, so $A_{hi} = m$
4	0.01544	0.02316	0.01930	0.02522	m ≤ F(m): solution above m, so $A_{lo} = m$
5	0.01930	0.02316	0.02123	0.02290	m ≤ F(m): solution above m, so $A_{lo} = m$
6	0.02123	0.02316	0.02219	0.02181	m > F(m): solution below m, so $A_{hi} = m$
7	0.02123	0.02219	0.02171	0.02235	m ≤ F(m): solution above m, so $A_{lo} = m$
8	0.02171	0.02219	0.02195	0.02208	m ≤ F(m): solution above m, so $A_{lo} = m$

7 Step 6: calibrating the APL gain

The gain a sets how strong APL is. It is chosen so that, with APL intact, 10% of KCs respond on average over odors, half the blocked level. More gain means more inhibition and fewer KCs responding, so the mean fraction falls steadily as a grows. The code (function `calib`) starts at $a = 1$ and doubles it until the fraction drops to 10% or below. It then bisects 50 times between the last two values. Every evaluation solves the fixed point of Step 5 for all odors.

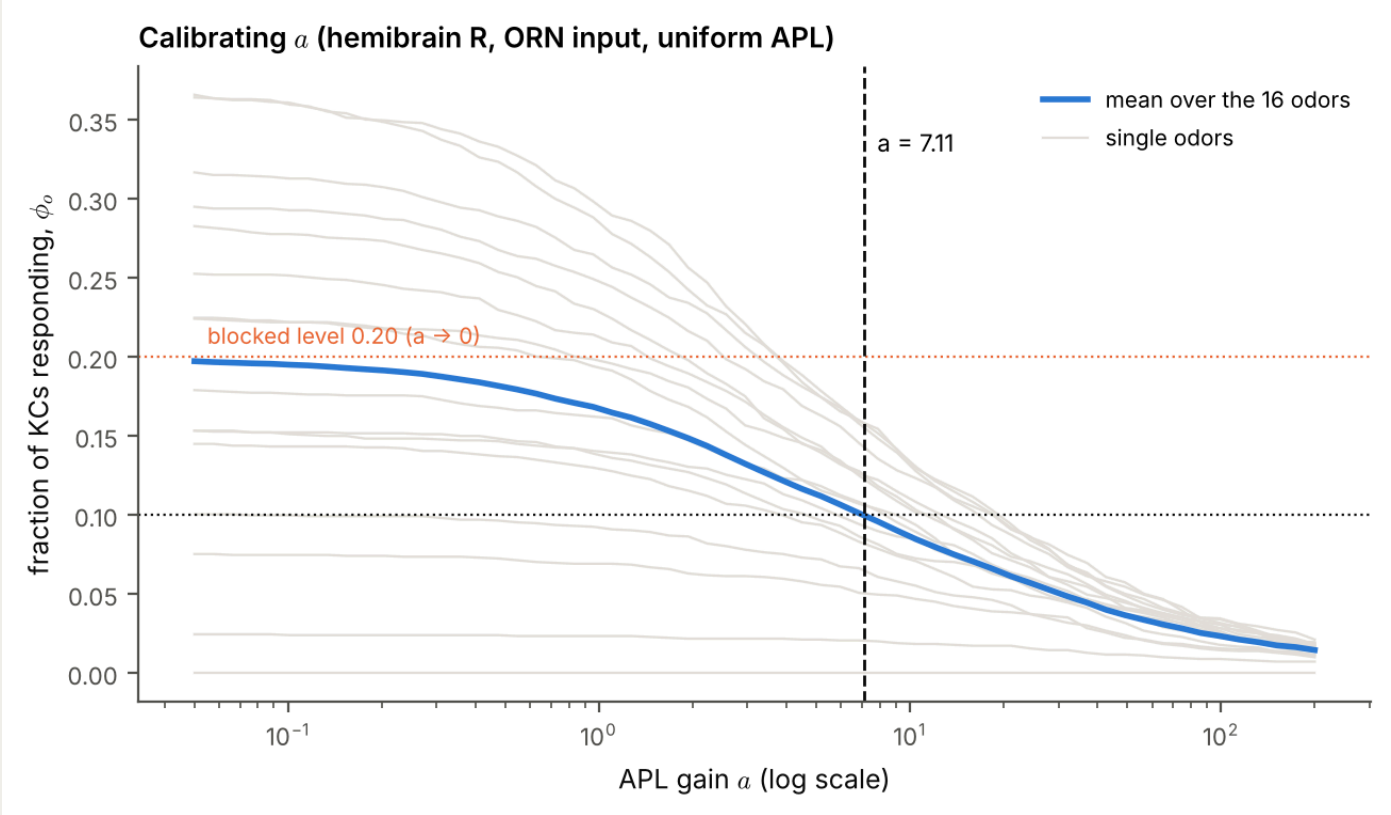


Figure 7. Calibrating the APL gain. The fraction of KCs responding with APL intact as the gain a grows (log axis). Thin grey lines are single odors, and the blue line is their mean. The calibrated gain (dashed) is where the mean reaches 0.10 (dotted black). At $a \to 0$ the model is the blocked model, with a mean of 0.20 (dotted orange).

Table 5. Calibrated gains (20% → 10%). U and C give a (subtractive); P0, P and L give b (divisive, on PN boutons or claws); M0 and M give the common scale s (Section 8). The gains are not comparable across columns, because each version uses a different form and normalization.

mushroom body	input	U	C	P0	P	L	M0	M
hemibrain R	ORN	7.11	10.6	20.9	19.7	21.9	0.977	0.971
male CNS L	ORN	7.11	14.5	20.5	20.4	20.4	0.972	0.97
male CNS R	ORN	7.24	10.6	20.5	21.2	20.8	0.973	0.978
hemibrain R	GH146	6.55	9.63	35.3	39.4	39.2	0.966	0.963
male CNS L	GH146	6.06	11.9	32.5	36.1	35.7	0.957	0.972
male CNS R	GH146	6.5	9.16	32.4	36.1	36.5	0.961	0.972

8 Step 7: what is read out

Three quantities are read out for every mushroom body, version and input table:

- **Fraction responding per odor**, ϕ_o, blocked and intact.
- **Odor-pair similarity** S_{ab}: the cosine similarity of the two odors' KC population response vectors $\mathbf{y}_a = (y_{1a}, \ldots, y_{Na})$ and $\mathbf{y}_b$ (Equation 8). It is the same measure as Yang's RDMs (which hold $1 - S$). Because responses are ≥ 0, S runs from 0 (no KC responds to both) to 1 (the same pattern up to a scale factor). It is undefined when an odor drives no KC.
- **The effect of blocking APL** on a pair, $\Delta S_{ab} = S_{ab}^{\text{blocked}} - S_{ab}^{\text{intact}}$ (Equation 9). It is computed per mushroom body and then averaged over the three.

$$S_{ab} = \cos(\mathbf{y}_a, \mathbf{y}_b) = \frac{\sum_k y_{ka}\, y_{kb}}{\sqrt{\sum_k y_{ka}^2}\,\sqrt{\sum_k y_{kb}^2}} \tag{8}$$

$$\Delta S_{ab} = S_{ab}^{\text{blocked}} - S_{ab}^{\text{intact}} \tag{9}$$

Odors that drive fewer than 0.5% of KCs with APL intact (in any version and mushroom body) are left out of the pair results, because their similarity rests on a handful of KCs. In the main setting this drops ms with ORN input and none with GH146 input.

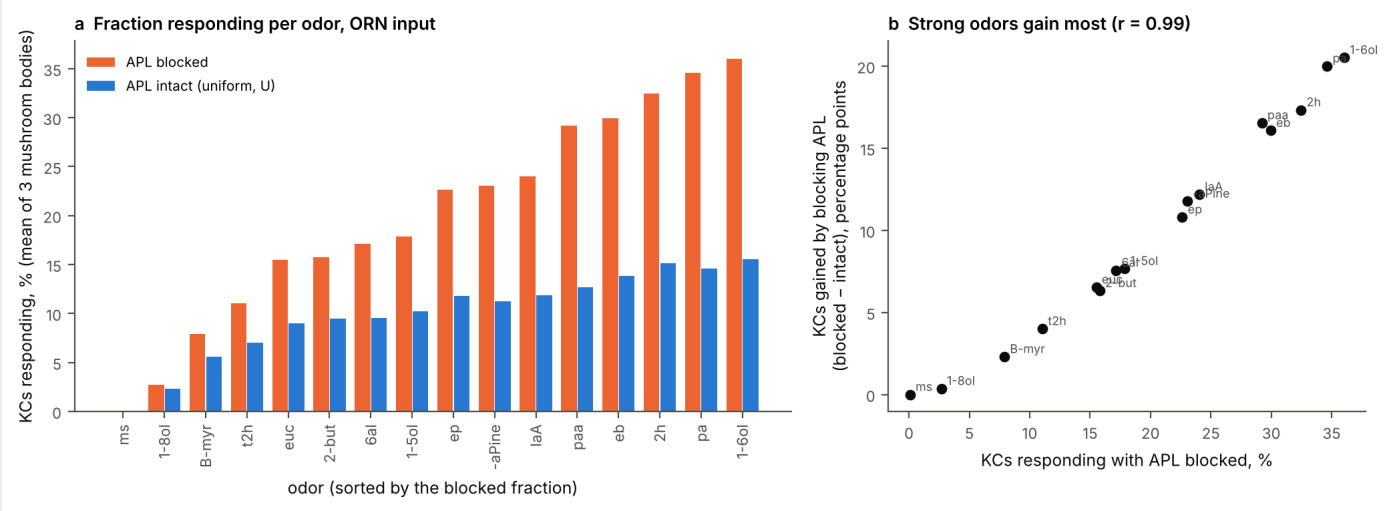


Figure 8. Per-odor fractions responding, blocked and intact. **a**, Percentage of KCs responding to each odor with APL blocked (orange) and intact (blue, uniform APL), averaged over the three mushroom bodies (ORN input, 20% → 10%). **b**, One point per odor. The horizontal axis is the blocked percentage (odor strength). The vertical axis is how many percentage points blocking APL adds. Strong odors recruit more APL, so removing APL adds the most KCs for them.

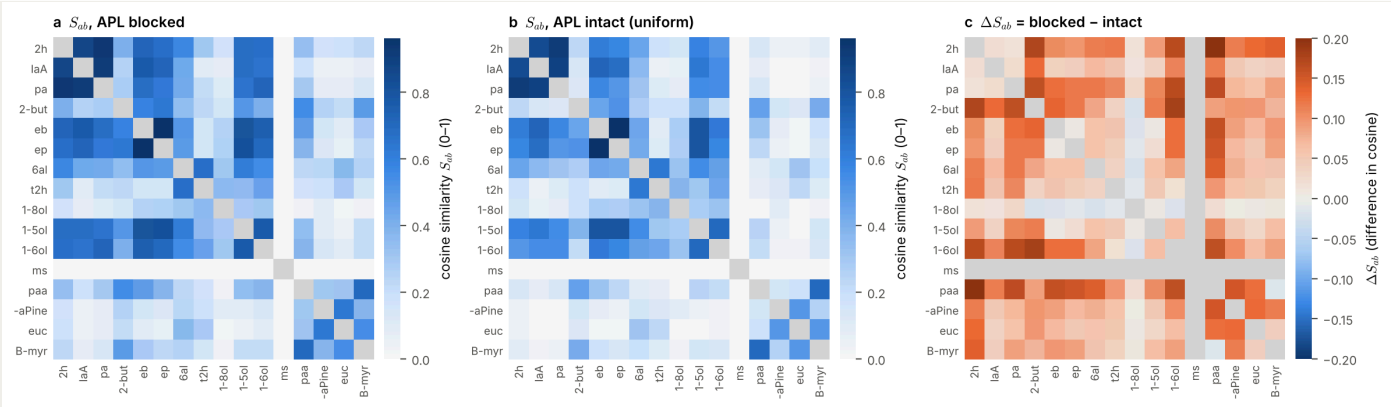


Figure 9. Odor-pair similarity with APL blocked and intact, and the change. Odor-by-odor matrices, mean of the three mushroom bodies, ORN input. **a**, **b**, Cosine similarity of the KC population responses with APL blocked and intact (uniform APL), on one shared scale (white 0, dark blue the highest value). **c**, Their difference, ΔS_{ab}, on a diverging scale: orange = more similar when APL is blocked, blue = less similar, grey-white = no change. Grey cells: diagonal, or an odor left out because fewer than 0.5% of KCs respond (ms). Mean ΔS = +0.076, range −0.02 to +0.20.

9 Step 8: versions that use APL's connectome wiring

Each version replaces part of the uniform APL with a connectome measurement. All of them keep the same blocked responses and the same θ, and all are calibrated to the same 10%. The versions therefore differ only in **where** the inhibition acts and **how it is weighted**. Table 6 summarizes them.

Table 6. The APL versions of the prediction run. a_U and b_{P0} are the calibrated gains of U and P0. In M0 and M only the common scale s is calibrated. Report 07 adds two feed-forward versions (F0, F), where APL activity comes from the PN input instead of the KCs.

version	KC response y_{ko}	APL activity A_o	weights from the connectome
U, uniform	$[x_{ko} - \theta - aA_o]_+$	$\frac{1}{N}\sum_k y_{ko}$	none ($u = v = 1$)
C, per-KC counts	$[x_{ko} - \theta - au_kA_o]_+$	$\frac{1}{N}\sum_k v_k y_{ko}$	u_k = APL→KC, v_k = KC→APL synapses of KC k in the calyx, each ÷ its mean
Cs, shuffled control	as C	as C	(u_k, v_k) pairs shuffled among KCs of the same class (αβ, α′β′, γ)
Cc, class means	as C	as C	u_k, v_k replaced by the mean of the KC's class
P0, boutons, all alike	$[\sum_g W_{kg}\frac{r_{go}}{1+bA_o} - \theta]_+$	$\frac{1}{N}\sum_k y_{ko}$	none
P, boutons per glomerulus	$[\sum_g W_{kg}\frac{r_{go}}{1+bp_gA_o} - \theta]_+$	as P0	p_g = APL→PN per PN→KC synapse of glomerulus g
L, claws per glomerulus	$[\sum_g W_{kg}\frac{r_{go}}{1+bq_gA_o} - \theta]_+$	as P0	q_g = on-claw APL→KC per claw PN synapse of g
M0, U + boutons at real share	$[\sum_g W_{kg}\frac{r_{go}}{1+sfb_{P0}A_o} - \theta - s(1-f)a_UA_o]_+$	as U	f = boutons' share of APL's calyx output
M, U + boutons per glomerulus	as M0 with p_g in the bouton term	as U	f and p_g

9.1 C: APL synapses per KC

u_k is KC k's number of APL→KC synapses in the calyx, divided by the mean over the mushroom body's KCs. v_k is the same for KC→APL synapses (calyx only since 2026-10-21, at Yang's request). A KC with $u_k = 2$ receives twice the average inhibition for the same APL activity, and a KC with $v_k = 2$ counts twice as much toward APL's activity. APL is still one unit with one activity per odor. The two controls ask whether it matters **which** KC has which counts (Cs keeps each KC class's distribution but shuffles it) or only the class averages (Cc).

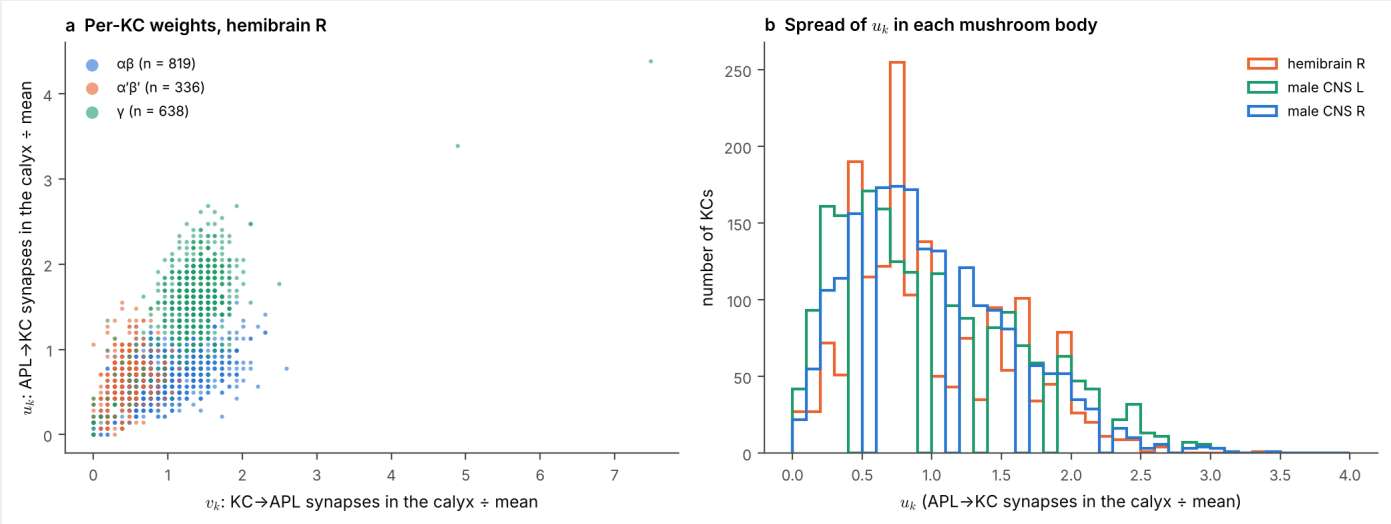


Figure 10. The per-KC weights used by version C. a, Each point is one hemibrain KC. Horizontal: KC→APL synapses in the calyx divided by the mushroom body mean (v_k). Vertical: APL→KC synapses divided by their mean (u_k). Colors are KC classes. **b**, Histograms of u_k in each mushroom body. Uniform APL corresponds to every KC sitting at (1, 1).

9.2 P, L, M: APL synapses per glomerulus

In P and L, inhibition **divides** a glomerulus's input instead of subtracting from the KC, because APL→PN synapses act presynaptically on the PN boutons and scale down what the boutons pass to all their KCs. p_g (P) is the number of APL→PN synapses on glomerulus g's boutons per PN→KC synapse of those boutons. q_g (L) is the number of APL→KC synapses on g's claws per PN synapse on those claws. Report 04 shows both across glomeruli. Within each mushroom body, the weights are scaled so that their claw-input-weighted mean is 1:

$$\tilde{p}_g = \frac{p_g}{\left(\sum_{g'} w_{g'}\, p_{g'}\right) / \sum_{g'} w_{g'}}, \qquad w_g = \sum_k W_{kg} \tag{10}$$

sums over glomeruli g' that have a value; glomeruli without a value get $\tilde{p}_g = 1$

With this scaling, P and P0 put the same total inhibition on the average claw input, and only its distribution across glomeruli differs. In M0 and M only a share f of the inhibition goes to boutons. f = APL→PN ÷ (APL→PN + APL→KC) synapses in the calyx, 0.097 in hemibrain and 0.112 in male CNS. The rest is uniform on the KCs. The KC gain is $(1 - f)\, a_U$, the bouton gain $f\, b_{P0}$, and both are scaled by one common factor s to reach 10%.

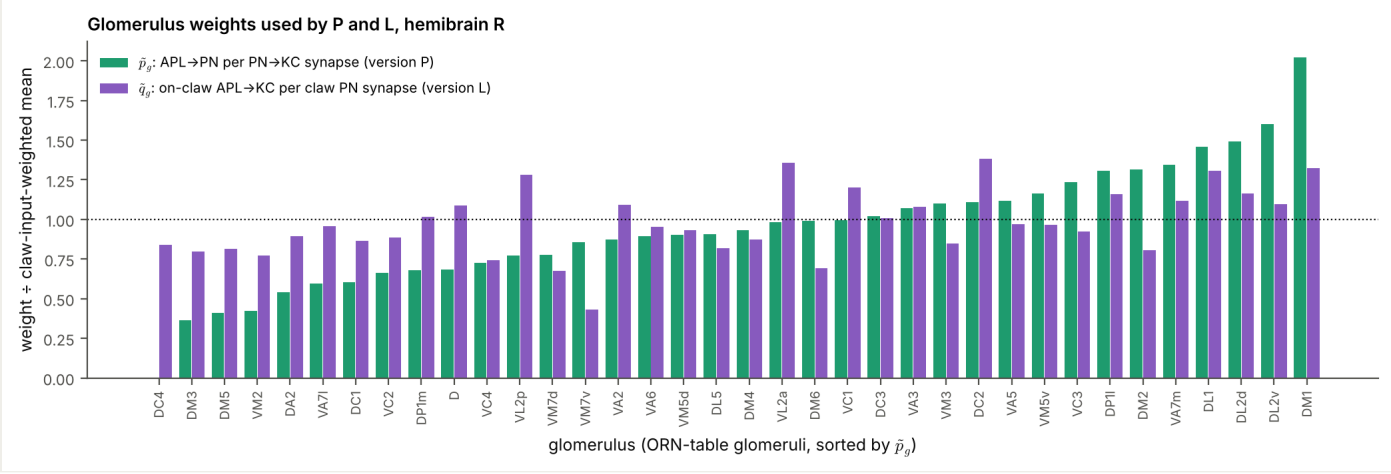


Figure 11. Per-glomerulus weights after scaling. For each glomerulus of the ORN table (hemibrain R), the bouton weight $\tilde{p}_g$ (green) and claw weight $\tilde{q}_g$ (purple), each divided by its claw-input-weighted mean (Equation 10), so the dotted line at 1 is the average. A glomerulus at 2 gets twice the average inhibition per unit of input in that version.

10 Results of the prediction run

- **Uniform APL.** The KCs gained by blocking APL track odor strength (r = 0.99 with ORN input), and so do all versions (r ≥ 0.999 between versions). Similarity rises most for pairs of strong odors: mean ΔS +0.076 with ORN and +0.041 with GH146. The pattern is the same in all three mushroom bodies (median r between them 0.87 and 0.81).
- **Per-KC counts (C).** They change ΔS by 0.029 on average (ORN), but the same counts shuffled within class do much the same (maps r 0.72). What is left differs between mushroom bodies (median r 0.34).
- **Boutons per glomerulus (P vs P0).** A consistent pattern (median r 0.87 between mushroom bodies; mean |change| 0.034): pa–eb and pa–ep rise more, eb and ep move away from 1-6ol and 1-8ol.
- **At the boutons' real share (M vs M0)** the same pattern shrinks to a mean |change| of 0.006.
- **Claws (L vs P0)**: mean |change| 0.018.

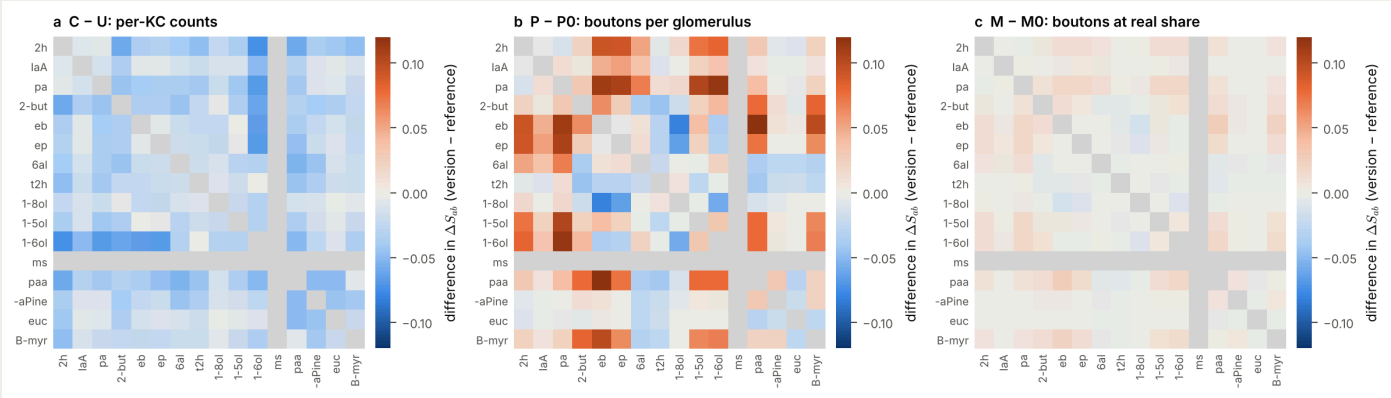


Figure 12. Wiring versions compared with their uniform references. How much each wiring version changes the effect of blocking APL, pair by pair (mean of three mushroom bodies, ORN input). Each cell is ΔS_{ab} of the version minus ΔS_{ab} of its reference. The blocked responses are identical, so this equals the reference's intact similarity minus the version's. Orange: blocking APL raises the pair's similarity more in this version. Blue: less. The scale is clipped at ±0.12. Grey: diagonal or an odor left out.

What this step does not include

One threshold for all KCs, with no differences in excitability between classes, and no spikes or time course.

APL is one unit with one activity per odor in every version. P and L weight its output per glomerulus but do not make its activity local. A local, non-spiking APL is modelled separately (`scripts/apl_local_model.py`).

The inputs are ORN or GH146 averages from other flies, not the recorded flies' PNs.

The coding levels (20% blocked, 10% intact) are assumptions here. Report 07 replaces them with the recordings.